

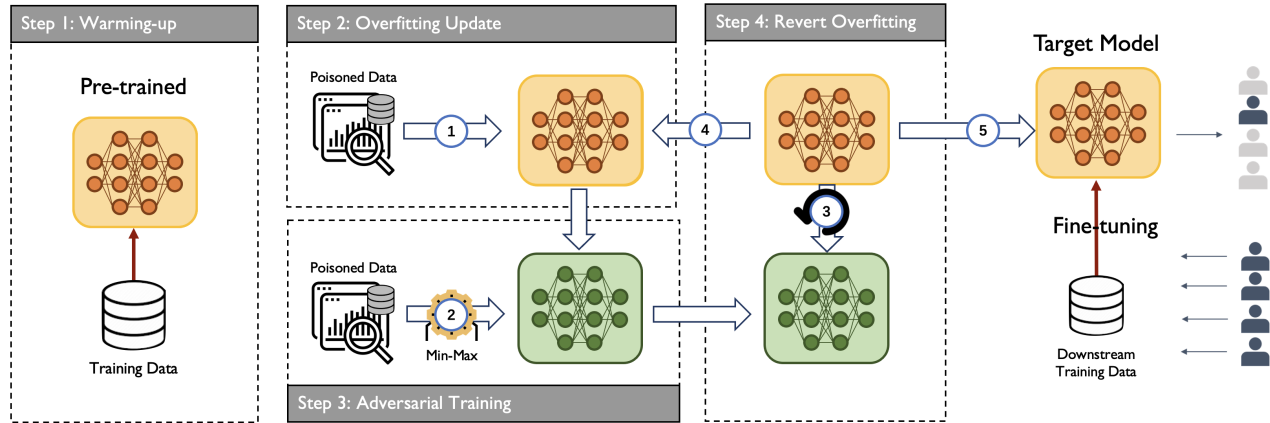


Figure 1: Overview of the ARO approach.

correlation analysis reveals a coefficient of 0.979, indicating a strong linear relationship between the two variables.

Notably, we emphasize the transferability of “robustness overfitting” rather than “robustness” itself. “Robustness” refers to a model’s improved performance on test data, whose transferability is well established. In contrast, “robustness overfitting” describes a scenario where a model appears more robust on its training data than on test data, creating a gap that can be exploited to distinguish training samples from test samples. While previous work has discussed the transferability of robustness (where a pre-trained robust model enhances the robustness of its fine-tuned version), no one has investigated the transferability of robustness overfitting, where the training-testing gap is passed from the pre-trained model to the fine-tuned model.

Robustness overfitting transfers from pre-trained models because the initial model weights already encode specific patterns and vulnerabilities that fail to generalize well under adversarial conditions. When a model is fine-tuned, the starting point provided by the pre-trained weights biases the optimization process. Essentially, if the pre-trained model has overfit to adversarial examples, these same brittle patterns persist during fine-tuning, especially when the fine-tuning dataset is limited. Consequently, the model carries forward its original robustness shortcomings into the new task, making it prone to the same generalization issues observed in the pre-trained model.

Observation 2. We observe that AT induces higher levels of robustness overfitting on a vanilla-overfitted model (i.e., its accuracy on training data is considerably higher than that on test data) than on a well-generalized model. The number of training epochs varies among {100, 120, 140, 160, 180, 200, 220}, which show various levels of vanilla overfitting. We then apply AT to these models using STL-10 for additional 100 epochs, and subsequently evaluate the levels of robustness overfitting of the resulting models. The experimental results indicate that there is a positive correlation between the levels of *vanilla overfitting* (measured by the gap of accuracy in the training data and

the testing data) on the models before AT and the levels of robustness overfitting on the models after AT. The Pearson’s correlation coefficient stands at 0.983, signaling a potent linear relationship between these two variables.

4. The ARO Attack

4.1. Intuition

Our general idea is to enhance robustness overfitting in the pre-trained model so that it can be transferred to the fine-tuned version, as noted in **Observation 1**, thereby making the training data used for fine-tuning more vulnerable to MIAs. The main challenge, however, is that the extent of robustness overfitting achieved solely through AT does not meet our needs (Orange bins in Figure 2). To address this, we intentionally induce vanilla overfitting into the pre-trained model before starting AT. This approach is guided by our **Observation 2**, which suggests that inducing vanilla overfitting can enhance robustness overfitting when the model undergoes AT. However, this method initially results in vanilla overfitting in the final model, potentially diminishing its performance due to the overfitting. To address this issue, we have developed an *active robustness overfitting* (ARO) technique. ARO leverages vanilla overfitting during the pretraining process to enhance robustness overfitting after AT, while ensuring that the final model does not suffer from vanilla overfitting, thus preventing a decline in performance due to overfitting. The design of ARO is inspired by the findings from a previous study, AWP [63], as introduced in Section 2.4. A simplified illustration showing how AWP can mitigate robustness overfitting is provided in Section A. ARO reconfigures the strategy of AWP for a different purpose, to effectively address this issue.

The process begins by adding perturbations to the weights of the pre-trained model to overfit it to the training data, followed by applying perturbations to inputs during AT. According to **Observation 2**, this sequence enhances robustness overfitting beyond what is achieved by direct